

Bachelor in Computer Engineering – Management and Information Systems

Introduction to Cybersecurity

Lab 3

Incident Response & Log Forensics

Student Assignment

Course	Introduction to Cybersecurity
Lab	3 of 4
Topic	SOC Analysis, Privilege Escalation, Lateral Movement
Duration	90–120 minutes
Difficulty	Intermediate

Scenario

HELIX Systems has suffered a security incident. The security team has detected suspicious activity during the early hours of 24 April 2026. You are a SOC analyst assigned to investigate the compromise. Your mission is to connect to each affected server, extract and analyse the logs, and reconstruct the full attack chain. Document your findings and identify the attacker's techniques.

Entry point: SSH to the lab gateway using the credentials provided by your instructor.

```
ssh analyst@lab-gateway-ip>
```

Password: An@l y s t 2024

Learning Objectives

- Read and interpret Linux authentication logs (/var/log/auth.log)
- Identify credential brute-force attacks from log evidence
- Recognise a sudo privilege escalation via find (GTF0Bins)
- Trace lateral movement between internal servers
- Document an incident timeline with supporting evidence
- Identify Indicators of Compromise (IOCs)

Lab Infrastructure

You have access to two servers inside the HELIX Systems network:

Host	IP	Role
helix-web	(via SSH gateway)	Web server – initial compromise
helix-db	192.168.60.10	Internal database server

Both servers have an analyst user account (An@l y s t 2024) for your investigation.

Tasks

Task 1 – Initial Access on helix-web (3 points)

Connect to helix-web and examine the SSH authentication logs.

```
grep "Failed password\|Accepted password\|Invalid user" /var/log/auth.log
```

Answer the following:

- a) From which external IP address did the attack originate?
- b) Which user account was targeted in the brute-force attack?
- c) At what time did the attacker successfully log in?

Task 2 – Privilege Escalation (4 points)

After obtaining initial access, the attacker escalated privileges.

```
grep "sudo" /var/log/auth.log | grep "COMMAND"  
cat /etc/sudoers.d/devops
```

Answer the following:

- a) What command did the attacker use to escalate privileges?
- b) What misconfiguration made this possible?
- c) What is the MITRE ATT&CK technique ID for this type of escalation?
- d) How could this misconfiguration have been prevented?

Task 3 – Persistence & Post-Exploitation (3 points)

Examine what the attacker did after gaining root access.

```
grep "useradd\|usermod\|passwd" /var/log/auth.log  
cat /etc/passwd | grep -v "nologin\|false"
```

Answer the following:

- a) What backdoor did the attacker create?
- b) What privileges does the backdoor account have?

Task 4 – Lateral Movement to helix-db (3 points)

Connect to the database server and look for signs of lateral movement.

```
ssh analyst@192.168.60.10  
grep "Accepted password" /var/log/auth.log
```

Answer the following:

- a) From which IP address did the attacker connect to helix-db?
- b) Which user account was used?
- c) At what time did this connection occur?

Task 5 – Data Exfiltration Evidence (3 points)

Examine the HELIX Systems data directory.

```
ls -la /opt/helix/data/  
cat /opt/helix/data/.exfil_marker
```

Answer the following:

- a) What sensitive data was present in this directory?
- b) What evidence suggests the data was accessed or exfiltrated?

Task 6 – Incident Report (4 points)

Write a brief incident report (200–300 words) containing:

- Executive summary of the incident
- Full attack timeline with timestamps
- List of Indicators of Compromise (IOCs)
- Recommended remediation steps

Submission

Submit a PDF document containing:

1. Answers to all tasks with supporting log evidence (screenshots or copy-paste)
2. The incident report from Task 6